



Lecture 6: Randomized/Probabilistic Algorithms

Example 1: Dataset Comparison

$$X = x_1x_2 \dots x_n, \quad x_i \in \{0,1\}$$

$$Y = y_1y_2 \dots y_n, \quad y_i \in \{0,1\}$$

Question: $X \stackrel{?}{=} Y$

Classical Communication Protocol:

Comparing the two datasets bit by bit:

- Communication overhead: $O(n)$
- Comparison overhead: $O(n) \Rightarrow n = 10^{16} \text{bits } (B) \approx 1136 \text{ TB}$

$$v(X) = \text{value}(X) = \sum_{i=1}^n x_i \cdot 2^{n-i}$$
$$\Pi(k) = \{p \in \mathbb{P} \mid p \leq k\}, \quad \pi(k) = |\Pi(k)|$$

Randomized Equal algorithm:

$$\text{Input: } \underset{A}{X = x_1x_2 \dots x_n}, \underset{B}{Y = y_1y_2 \dots y_n}, \quad x_iy_i \in \{0,1\}$$

1. A randomly chooses a prime number $p \in \Pi(n^2)$
2. A calculates the “fingerprint of $X \Rightarrow s = v(X) \bmod p$.
 - a. **Note:** X is considered as a binary number
3. A sends s and p to B
4. B calculates the “fingerprint of $Y \Rightarrow t = v(Y) \bmod p$
5. B compares whether $s = t$? Are the two fingerprints the same?
 - a. Yes: equal $\rightarrow A$,
 - b. No: unequal $\rightarrow A$

Comparison overhead:

The Equal algorithm is a randomized communication protocol for the data comparison example. It drastically reduces the comparison effort. Comparison overhead was previously $O(n)$, now the overhead is reduced to $5 \rightarrow O(1)$



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Communication overhead:

$$0 \leq p, s \leq n^2$$

$$m \in \mathbb{N} \rightarrow \lceil \log m \rceil \text{ Bits}$$

$$l(s, p) \leq 2 * \lceil \log n^2 \rceil \leq 4 * \lceil \log n \rceil$$

For $n = 10^{16}$;

$$\leq 4 * 16 * \lceil \log 10 \rceil$$

$$\leq 4 * 16 * 4$$

$$= 256 \text{ Bits}$$

Instead of 10^{16} , we would only transmit 256 bits.

$$Prob_{equal}[A | B]$$

$$X = Y \Rightarrow t = s$$

$$X \neq Y \Rightarrow t \neq s$$

$$Prob_{equal}["unequal" | X = Y] = 0$$

Example:

$$n = 5$$

$$X = 10011 \quad v(X) = 19$$

$$Y = 10001 \quad v(Y) = 17$$

$$\text{for } p = 11 \Rightarrow s = 8, t = 6$$

$$X = 10011 \quad v(X) = 19$$

$$Y = 10001 \quad v(Y) = 17$$

$$\text{for } p = 2 \Rightarrow s = 1, t = 1$$

$p = 2$ is a bad witness for the equality of X and Y

Question: How many bad witnesses are in $\Pi(n^2)$?

$s = t$, i.e. $v(X) = v(X) \bmod (p)$, although $X \neq Y$.

$$\Pi^+(n^2, X, Y) = \{p \in \mathbb{P} \mid v(X) \neq v(Y)(p), X \neq Y\}$$

$$\Pi^-(n^2, X, Y) = \{p \in \mathbb{P} \mid v(X) = v(Y)(p), X \neq Y\}$$



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Establishing the relationship between bad witnesses and total witness candidates:

$$Prob_{equal}[X = Y \mid X \neq Y] = \frac{|\Pi^-(n^2, X, Y)|}{|\Pi(n^2)|} = \frac{2 \ln n}{n}$$

$$\text{for } n = 10^{16} \Rightarrow \frac{2 \ln 10^{16}}{10^{16}} \approx 0.7 * 10^{-14}$$

Multiple rounds of algorithm to further reduce the error:

$$O(l * \log n)$$

$$l = 10 \Rightarrow 0.7 * 10^{-144}$$

$$L \subseteq \Sigma^*, w \in \Sigma^*: w \in^? L$$

$$D = \{(x, y) \in \{0,1\}^n \times \{0,1\}^n \mid n \in \mathbb{N}_0, x \neq y\}$$

Given $w \in \{0,1\}^n \times \{0,1\}^n$

1. $Prob_{equal}["w \in D" \mid w \notin D] = 0$
2. $Prob_{equal}["w \in D" \mid w \in D] \geq 1 - \frac{2 \ln n}{n} \Rightarrow RP(\varepsilon(n)), \varepsilon(n) \text{ is the error bound}$
3. Overhead: $O(\log n)$

Example 2: Triangle Graph

$$\Delta Graph = \{\langle G \rangle \mid G \text{ undirected graph that contains at least one triangle}\}$$

1. T (tester) randomly chooses an edge $\{a, b\}$
2. T randomly chooses a node $c \neq \{a, b\}$
3. Test: Do c and $\{a, b\}$ form a triangle? -> Yes or No

$$Prob_T["yes" \mid G \notin \Delta Graph] = 0 \rightarrow \text{no false positive statement}$$

$$G = (V, E)$$

$$|V| = n, |E| = m$$

$$Prob_T["yes in one round" \mid G \in \Delta Graph] \geq \frac{3}{m} * \frac{1}{n-2}$$



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$$Prob_T["no in one round" \mid G \in \Delta Graph] \leq 1 - \frac{3}{m(n-2)}$$

$$Prob_T["no in l rounds" \mid G \in \Delta Graph] \leq \left(1 - \frac{3}{m(n-2)}\right)^l$$

$$Prob_T["yes in l rounds" \mid G \in \Delta Graph] \geq 1 - \left(1 - \frac{3}{m(n-2)}\right)^l$$

$$\left(1 + \frac{1}{k}\right)^k \xrightarrow{k \rightarrow \infty} e^1 = 2.7182 \dots$$

More general formulation with x instead of the special case with 1:

$$\left(1 + \frac{x}{k}\right)^k \xrightarrow{k \rightarrow \infty} e^x$$

Place a minus before x:

$$\left(1 + \frac{-x}{k}\right)^k \xrightarrow{k \rightarrow \infty} e^{-x}$$

We want to apply the Euler sequence for our probabilistic $\Delta Graph$ algorithm:

$$k = l, x = yl \rightarrow (1 - y)^l \xrightarrow{l \rightarrow \infty} e^{-yl}$$

$$y = \frac{3}{m(n-2)}, l = \frac{m(n-2)}{3} \rightarrow y * l = 1$$

$$\left(1 - \frac{3}{m(n-2)}\right)^{\frac{m(n-2)}{3}} \approx e^{-1} \approx \frac{1}{e} \approx \frac{1}{2.7} < \frac{1}{2}$$

RP

1. $Prob_T["yes" \mid G \notin \Delta Graph] = 0$
2. $Prob_T["yes" \mid G \in \Delta Graph] \geq \frac{1}{2}$
3. *polynomial*

Outline:

$$Dataset \in RP\left(\frac{2 \ln n}{n}\right)$$

$$\Delta Graph \in RP\left(\frac{1}{2}\right)$$



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Complexity Class RP:

Random/Probabilistic polynomial running algorithms with one-sided error

- No false positive statements.
- Conversely, RP algorithms make errors with a particular bound ε .
- By repeating the execution of the algorithm, the total error can be reduced.

⇒ The algorithms that class RP defines are also called **Monte Carlo algorithms**. Such algorithms allow a one-sided error.